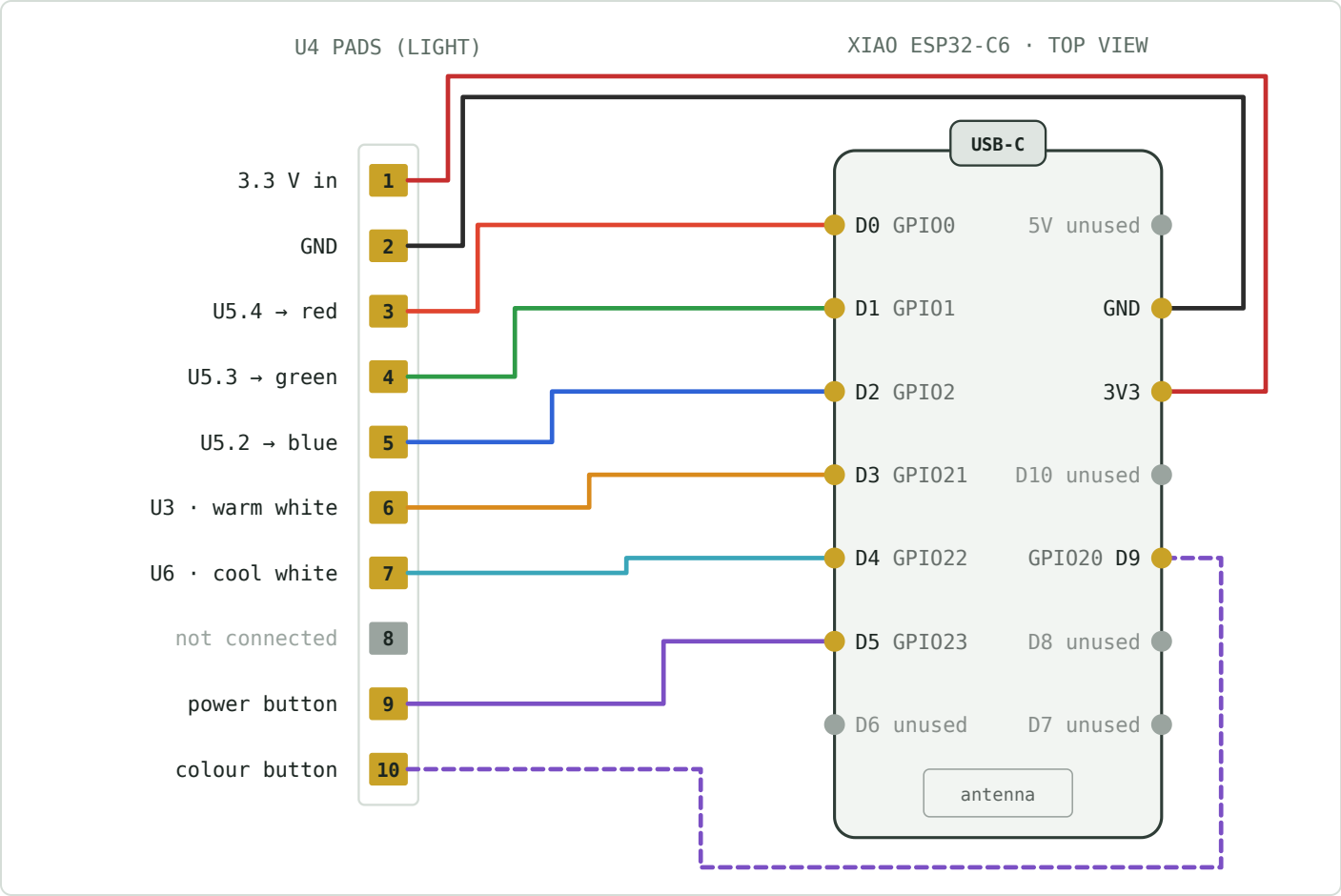
CYNC REVEAL HD+ LIGHT · U4 REPLACEMENT

An ESP32-C6 in place of the JXC8720-18

Remove the Wi-Fi/BLE module at U4 and wire a Seeed Studio XIAO ESP32-C6, or an Adafruit Feather ESP32-C6, to its ten pads. The light's three SM15133EH drivers, the buttons and the 3.3 V regulator stay as they are. Pin choices match the ESPHome package in `RoussLabs/ESPHomeGeCyncESP32Modification`.

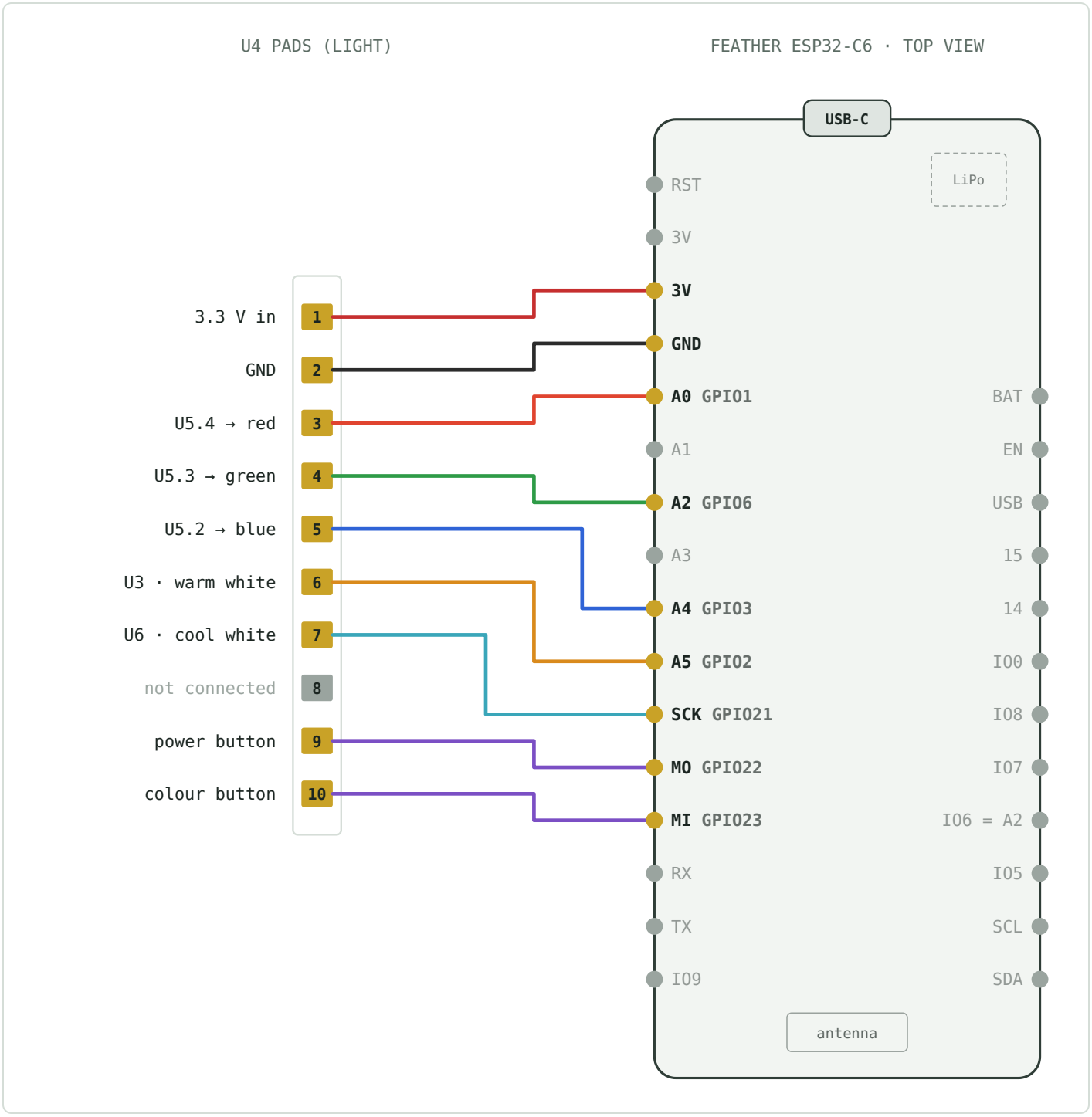
Wiring: Seeed Studio XIAO ESP32-C6



Pad numbers are as traced on the light: pad 1 is 3.3 V, pad 2 is GND. The dashed wire runs under the XIAO to reach D9 on the far header. The XIAO is the recommended board: at 21 × 17.5 mm it fits where the old module was.

Wiring: Adafruit Feather ESP32-C6

The Feather works with the same firmware, with different pins. Every wire goes to its long header, in the same order as the U4 pads, so nothing crosses. Nothing connects to the short header.



Wire by the labels printed on the Feather. A2 is also brought out as IO6 on the short header; leave that pin empty. The pins drawn as unused include the onboard red LED (15) and the BOOT button and NeoPixel (IO9).

Config for a Feather light

Add these to that light's own YAML (the file made from `examples/undercabinet-light.yaml`). Values in the device file override the package, so everything else stays shared.

```
substitutions:
  name: undercabinet-light-01
  friendly_name: Undercabinet Light 01
  # Feather ESP32-C6 pins
  red_pin: GPIO1            # A0
  green_pin: GPIO6          # A2
  blue_pin: GPIO3           # A4
  warm_white_pin: GPIO2     # A5
  cool_white_pin: GPIO21    # SCK
  power_button_pin: GPIO22  # MO
  colour_button_pin: GPIO23 # MI

esp32:
  board: adafruit_feather_esp32c6
```

Feather notes

- **Size.** The Feather is about 51 × 23 mm, against 21 × 17.5 mm for the XIAO. Check it fits inside the light's housing before choosing it. It will not sit where the module was, so mount it clear of the board and keep the wires as short as you can.
- **Power.** The light's 3.3 V feeds the Feather's 3V pin, the output of its own regulator. Adafruit notes that feeding 3V directly bypasses the regulator, so the EN pin no longer works; that is fine here. Leave BAT, USB and EN unconnected.
- **Pins.** None of the seven GPIOs are ESP32-C6 strapping pins (GPIO4, 5, 8, 9, 15), so A1 (GPIO4), A3 (GPIO5), IO8, IO9 and 15 stay free. RX and TX (UART0) are left free too.

Wire list

WIRE	U4 PAD	REACHES ON THE LIGHT	XIAO PIN · GPIO	FEATHER PIN · GPIO	CONFIG SUBSTITUTION
Power	1	3.3 V from the light's regulator	3V3	3V	–
Ground	2	Ground	GND	GND	–
Red	3	U5.4 DIM, red LEDs (154 Ω in series)	D0 · GPIO0	A0 · GPIO1	red_pin
Green	4	U5.3 DIM, green LEDs	D1 · GPIO1	A2 · GPIO6	green_pin
Blue	5	U5.2 DIM, blue LEDs	D2 · GPIO2	A4 · GPIO3	blue_pin
Warm white	6	U3.2/3/4 DIM, all three channels	D3 · GPIO21	A5 · GPIO2	warm_white_pin
Cool white	7	U6.2/3/4 DIM, all three channels	D4 · GPIO22	SCK · GPIO21	cool_white_pin
Power button	9	Power button, 0 V when pressed	D5 · GPIO23	MO · GPIO22	power_button_pin
Colour button	10	Colour button, 0 V when pressed	D9 · GPIO20	MI · GPIO23	colour_button_pin
No wire	8	Not connected on the light	–	–	–

The XIAO pins are the package defaults; a Feather light overrides them as shown above. On the XIAO, D6 and D7 (UART0), D8, D10 and the antenna-switch pins (GPIO3, GPIO14) are left alone.

Fit-up and first power-on

> **No USB while wired.** The board's own regulator and the light's 3.3 V supply would both drive the same rail. Flash it on the bench over USB first, then install it and update over the air.

- [ ] Flash the light's YAML over USB with the board off the light, and check it joins Wi-Fi and appears in Home Assistant.
- [ ] Remove the JXC8720-18 from U4 with hot air and clean the pads.
- [ ] Insulate the underside of the board with Kapton tape. Its battery pads and test points must not touch the light's board.
- [ ] Keep the wires short. Put the board's antenna end where the old module's antenna was, at the board edge and away from the metal channel.
- [ ] On first power-up, watch for any channel that flashes during boot. If one does, add a 100 kΩ pull-down from that line to GND.
- [ ] Set the light to pure red, green and blue in Home Assistant and check each colour. The pin order was confirmed on the first light.
- [ ] Dim to 1% and raise `Minimum Brightness` until every channel glows evenly; it uses the same % as the brightness slider. `PWM Frequency` defaults to 4882 Hz, which showed no flicker on the first light.

References

1. RoussLabs/ESPHomeGeCyncESP32Modification · this project's ESPHome package, example config and component
2. Getting Started with Seeed Studio XIAO ESP32C6 · Seeed Studio wiki: pinout, power, antenna
3. XIAO ESP32C6 pin multiplexing · Seeed Studio wiki
4. Adafruit ESP32-C6 Feather · Adafruit Learning System: pinouts and power pins
5. Adafruit-ESP32-C6-Feather-PCB · Adafruit: schematic, board files and PrettyPins pinout
6. adafruit_feather_esp32c6 pins_arduino.h · Espressif arduino-esp32: A0–A5, SPI and UART GPIO numbers
7. ESP32-C6 Hardware Design Guidelines · Espressif: strapping pins and power
8. ESP32 LEDC Output · ESPHome: PWM frequency and resolution
9. Packages · ESPHome: shared config and overrides in the device file
10. Security best practices · ESPHome: one API encryption key per device
11. FCC ID PUU-CFIXCNLRCRV · the Cync reveal HD+ undercabinet light's FCC filing